SCTP ASSOCIATE DATA ANALYST CAPSTONE PROJECT

IBM HR Employee Attrition Analytics

From Data to Retention Decisions

Understanding Attrition • Predicting Risk • Supporting HR Retention Strategies



Capstone Project Presentation

November 2026

Python | Power BI | Streamlit

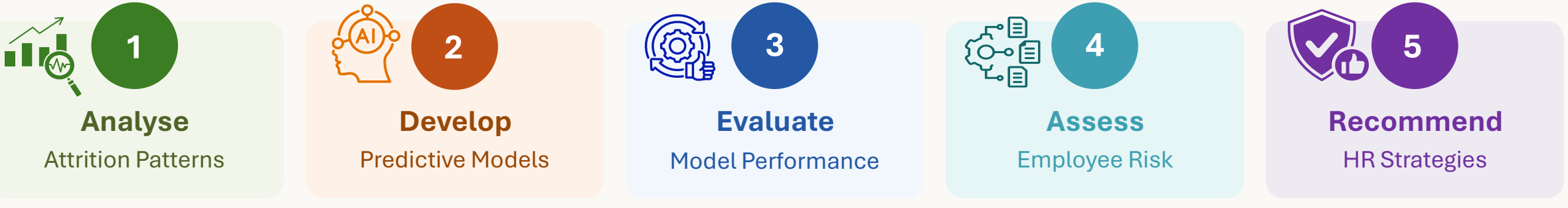
BUSINESS PROBLEM & PROJECT OBJECTIVES

Business Problem

How can employee data be analysed to identify key factors associated with employee attrition, predict employees' likelihood of leaving, and support targeted retention strategies and proactive workforce planning to reduce attrition and minimise staffing gaps?

Project Objectives

1. Analyse employee attrition patterns across departments, job roles and key employee characteristics to identify groups with higher attrition rates.
2. Develop and compare machine learning models to predict employee attrition.
3. Evaluate the selected predictive model, assess probability calibration and identify important predictors of attrition.
4. Assess predicted employee attrition risk and develop an interactive Power BI dashboard to support workforce-level analysis and employee risk investigation.
5. Recommend practical HR retention strategies and an implementation plan to help reduce attrition and support proactive workforce planning.



PROJECT FLOW | Understand → Predict → Validate → Investigate → Act

DATA UNDERSTANDING

The IBM HR Analytics dataset contains employee demographics, compensation, performance, work experience and employment status information.

These datasets support workforce analysis, identification of attrition patterns and data-driven HR retention strategies.

Data Period: Not specified in the source dataset documentation.

Source Table	Purpose & Key Variables
Employees	Employee demographics and job profiles, including age, gender, department and job role.
Compensation	Employee salary and rewards information, including monthly income, salary increments and stock options.
Performance	Employee performance, satisfaction, engagement and work-life balance indicators.
Experience	Employee career history, tenure, training and promotion progression.
Employment Status	Employee attrition and overtime information used for retention analysis and predictive modelling.

Target Variable: Attrition (Yes/No)

DATA PIPELINE

From Raw Data to Actionable Insights

Five source datasets feed a reproducible analysis and two interactive outputs.

5
Source Datasets

1,470
Employee Records

22
Model Predictors

Source Datasets

Employees
Employee demographic & profiles

Performance
Performance related measures

Employment Status
Employee attrition outcomes

Compensation
Salary & reward information

Experience
Work history & experience

Google Colaboratory

python

Merge on Employee Number → Clean & Prepare → Analyse & Model

Power BI
Interactive HR dashboards

Streamlit
Live prediction and risk exploration

DESCRIPTIVE ANALYTICS

1

Analyse

Attrition Patterns

Understanding Attrition

1,470	1,233	237	16.1%
Total Employees	Employees Stayed	Employees Left	Attrition Rate

Attrition Rate by Department



KEY INSIGHT

Sales recorded the highest attrition at 20.6%, exceeding the overall rate of 16.1% by 4.5 percentage points.

DESCRIPTIVE ANALYTICS

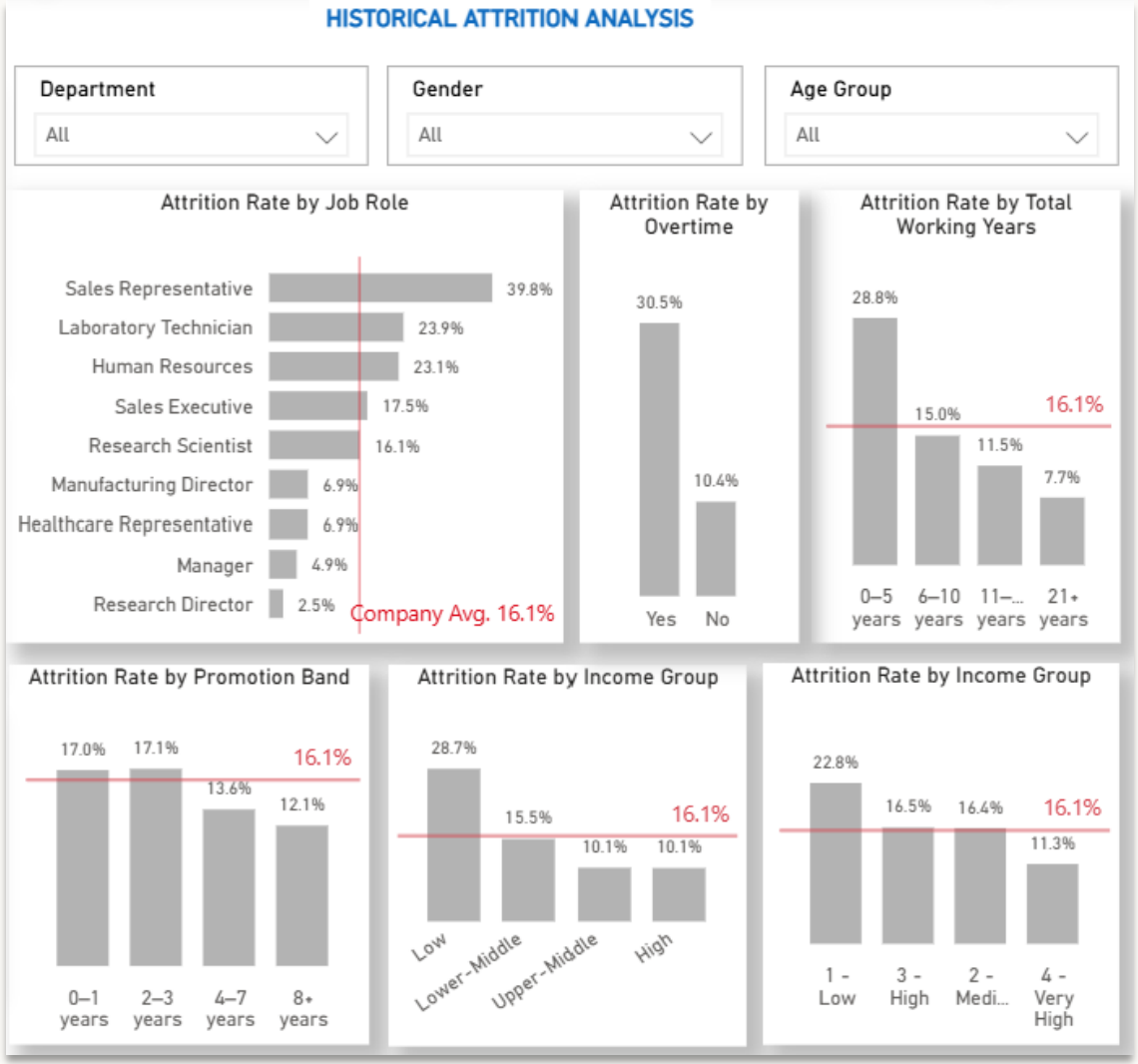
1

Analyse

Attribution Patterns

Understanding Attrition Beyond Departments

Multiple factors provide a more complete view of employee attrition.



Employee Profile

Age, job role and experience

Work Characteristics

Overtime, business travel and workload indicators

Career & Engagement

Promotion history, job satisfaction and career progression

ANALYTICAL APPROACH

Examine multiple employee characteristics to better understand attrition patterns, rather than relying on a single factor.

MODEL DEVELOPMENT & EVALUATION

2 **Develop**
Predictive Models

3 **Evaluate**
Model Performance

Logistic Regression Delivered Stronger Predictive Performance

● Logistic Regression ● Random Forest | 5-fold cross-validation

ROC AUC



Recall

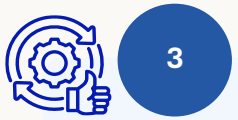


Selected: Calibrated Logistic Regression

0.795	72.3%	34 / 47
Test ROC AUC	Test Recall	Actual Leavers Identified

KEY TAKEAWAY | Higher recall and ROC AUC supported Logistic Regression. Calibration improved probability estimates.

PREDICTOR IMPORTANCE



3

Evaluate
Model Performance

What Matters Most in Predicting Attrition?

Top five predictors from permutation importance analysis.

1	**Overtime**	Work arrangements
2	**Stock Option Level**	Employee rewards
3	**Years Since Last Promotion**	Career progression
3	**Number of Companies Worked**	Employment history
4	**Total Working Years**	Work experience

HR INTERPRETATION | Work arrangements, rewards, career progression and experience are useful areas for further investigation.

Calibration improved cross-validated Brier Score from 0.160 to 0.098 (lower is better). Predictor importance does not establish causality.

EMPLOYEE RISK ASSESSMENT

4

Assess
Employee Risk

134 Employees Identified as High Risk

Predicted attrition risk among 1,233 employees who stayed.



Low Risk	Medium Risk	High Risk
733	366	134
59.5%	29.7%	10.9%

Workforce-Level View

Understand how predicted risk is distributed across employees and departments.

Employee-Level View

Investigate employee risk profiles to support further HR assessment.

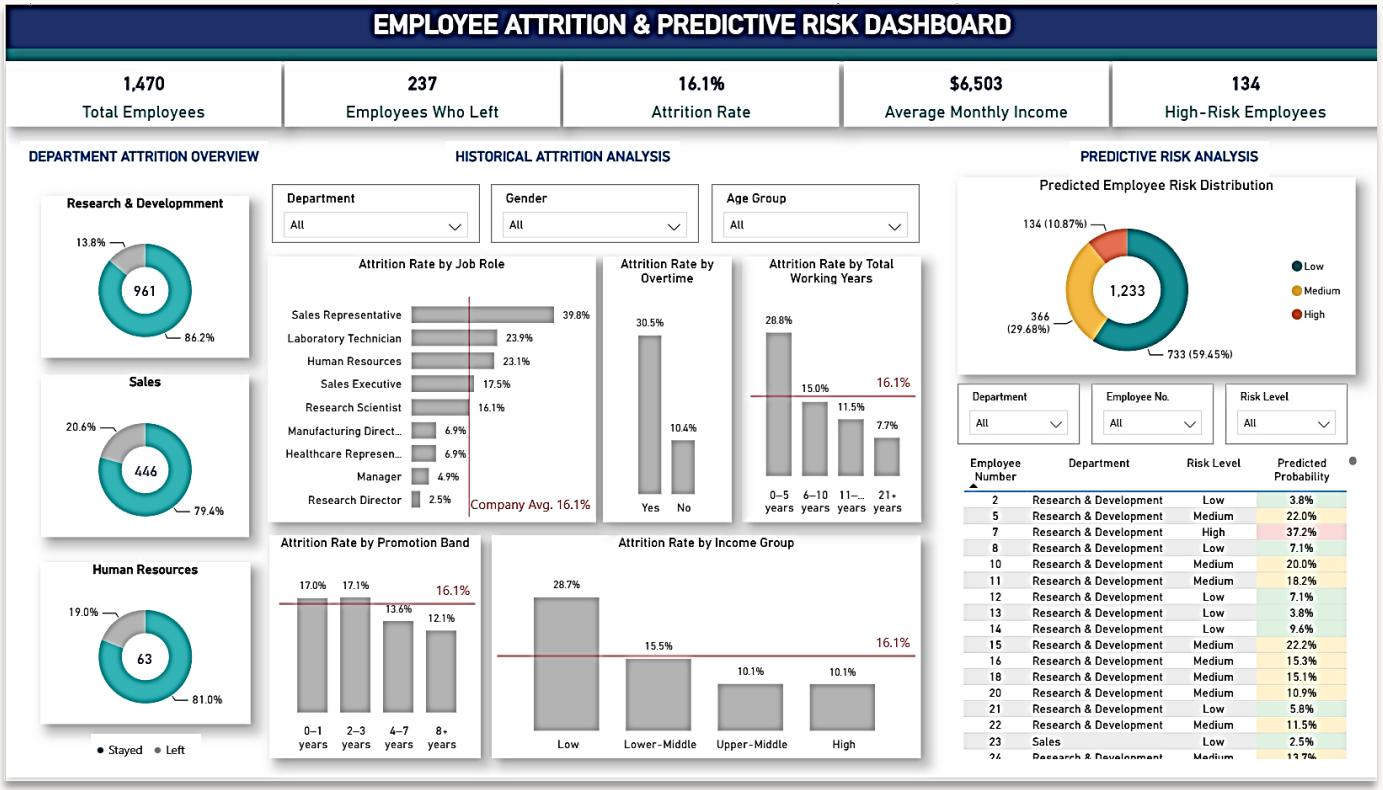
KEY TAKEAWAY | 10.9% of employees who stayed were classified as high risk, providing HR a starting point for investigation.

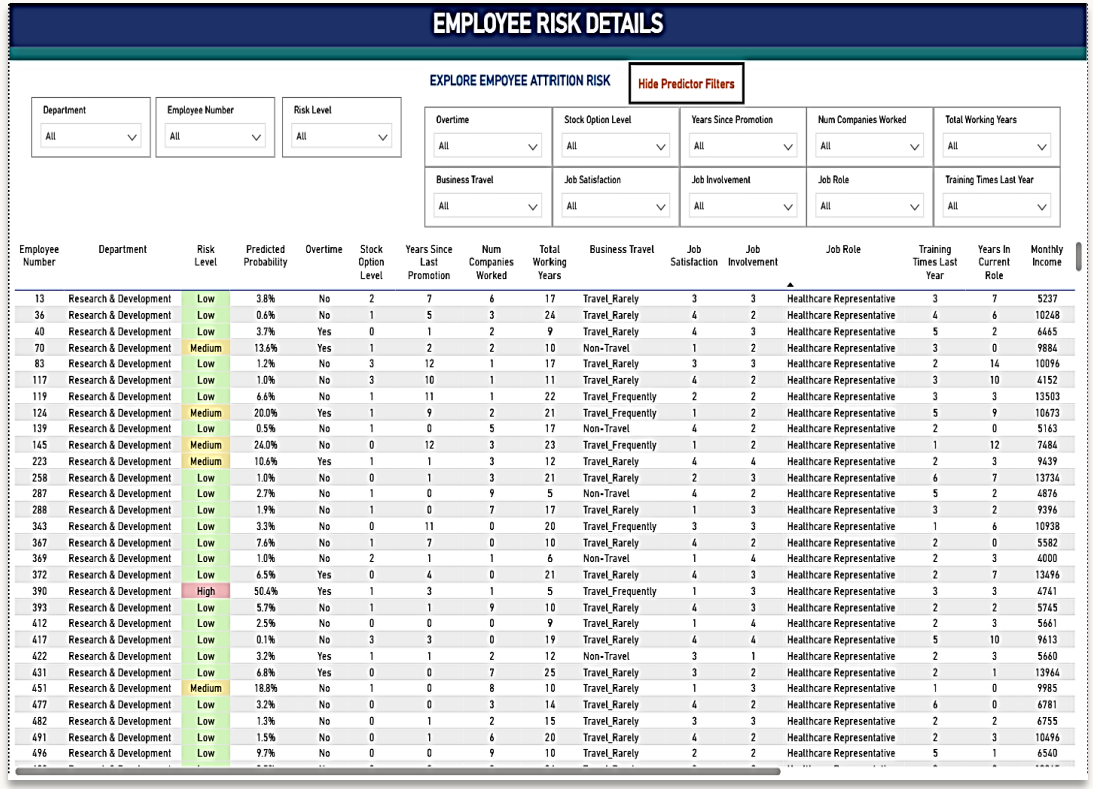
Model-estimated risk using synthetic IBM HR data; not confirmed future resignations.

INTERACTIVE ANALYTICS

Power BI Dashboards

Two complementary views: workforce attrition and employee risk details.

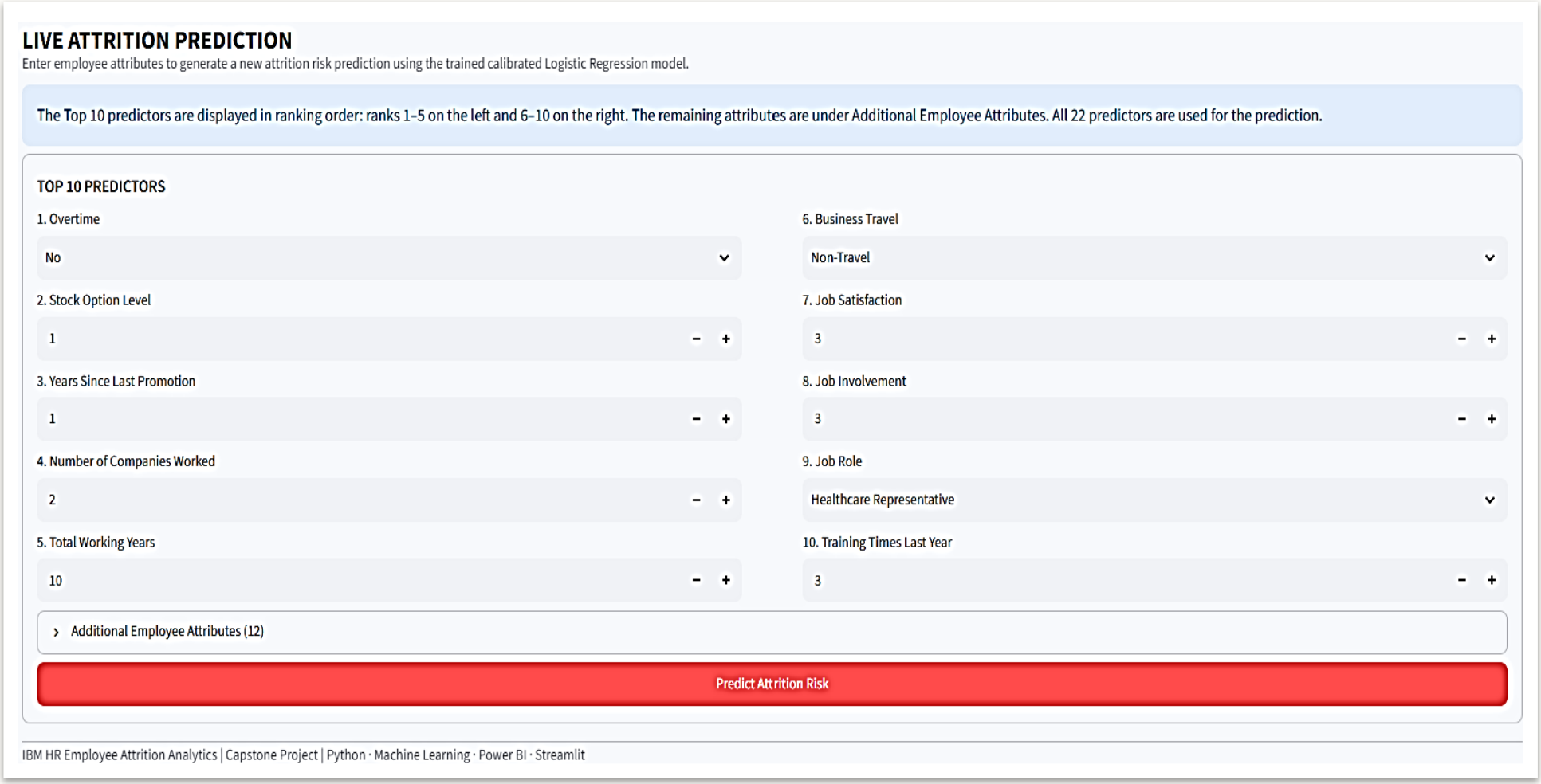




STREAMLIT APPLICATION – LIVE PREDICTION

Predicting Attrition Risk in Real Time

Employee characteristics can be adjusted to generate a new model-estimated attrition probability.



1. Enter employee characteristics
2. Generate predicted risk
3. Support HR investigation

HR RETENTION STRATEGIES

5

Recommend
HR Strategies

From Predictive Insights to Targeted Retention Strategies

Four practical areas for HR to investigate and address.

01 **Workload & Well-being**

Review overtime patterns, workload distribution and employee well-being.

02 **Career Development**

Strengthen career conversations, development plans and progression opportunities.

03 **Rewards & Recognition**

Review employee rewards, recognition and retention incentives.

04 **Targeted HR Engagement**

Prioritise supportive follow-ups and further assessment of higher-risk groups.

RECOMMENDED APPROACH | Use model findings to guide supportive HR interventions, not employment decisions based solely on risk.

IMPLEMENTATION & CONCLUSION

5

Recommend
HR Strategies

From Prediction to Prevention

A phased approach to translate attrition analytics into practical HR decision support.

01 Validate

Test the model with real HR data and review fairness, calibration and performance.

02 Pilot

Introduce risk insights to HR teams and conduct supportive employee engagement.

03 Monitor

Track attrition trends, interventions and model performance over time.

04 Refine & Scale

Improve the approach based on pilot outcomes before wider implementation.

FINAL TAKEAWAY | Turning employee data into earlier, more informed retention decisions.

Proof of concept using synthetic IBM HR data — real-world validation is required before operational use.

Thank You

From Data to Retention Decisions

Questions & Discussion